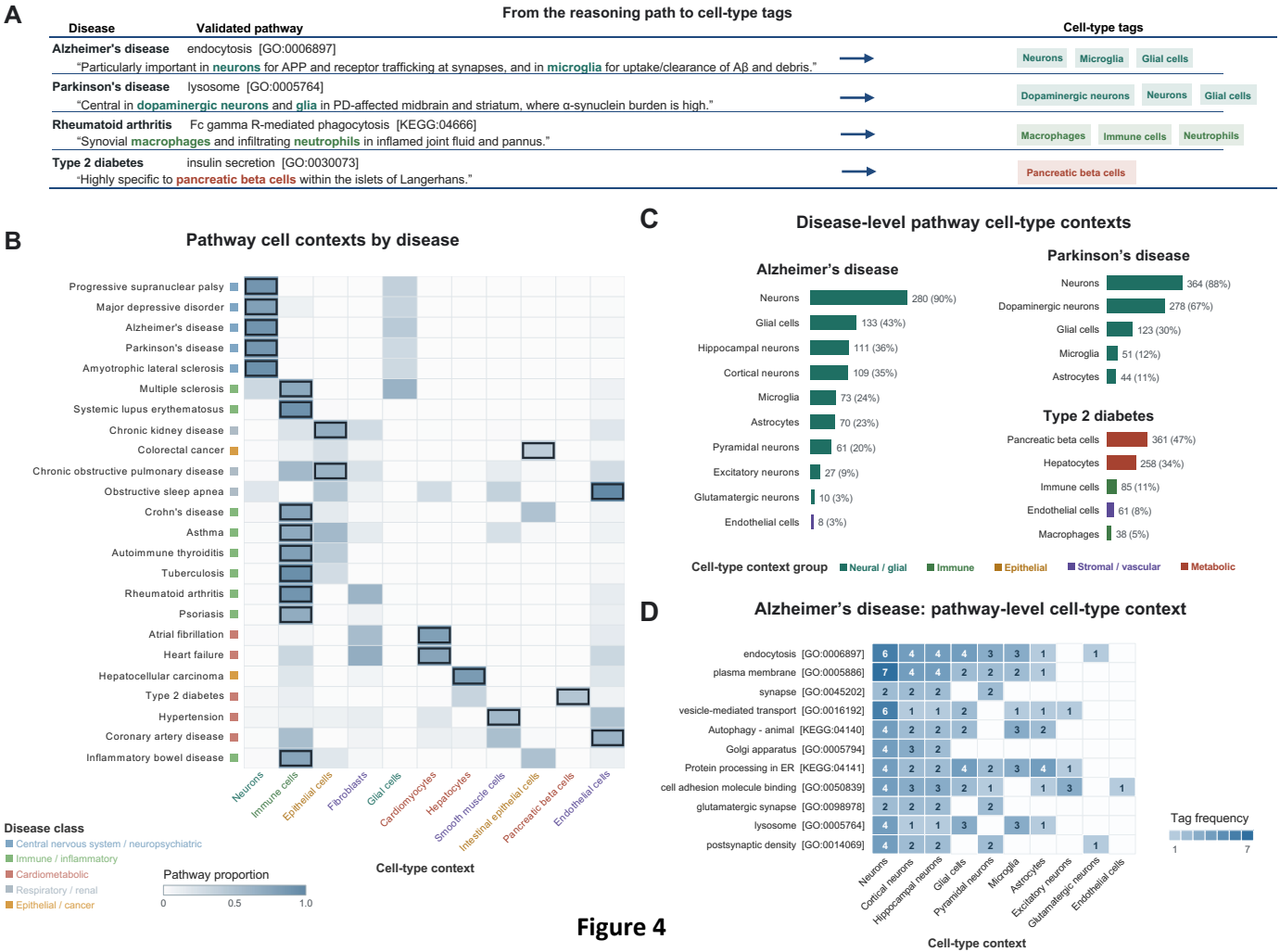

A
From the reasoning path to cell-type tags
Disease
Validated pathway
Cell-type tags
Alzheimer's disease
endocytosis [GO:0006897]
"Particularly important in neurons for APP and receptor trafficking at synapses, and in microglia for uptake/clearance of Aβ and debris."
Neurons
Microglia
Glial cells
Parkinson's disease
lysosome [GO:0005764]
"Central in dopaminergic neurons and glia in PD-affected midbrain and striatum, where α-synuclein burden is high."
Dopaminergic neurons
Neurons
Glial cells
Rheumatoid arthritis
Fc gamma R-mediated phagocytosis [KEGG:04666]
"Synovial macrophages and infiltrating neutrophils in inflamed joint fluid and pannus."
Macrophages
Immune cells
Neutrophils
Type 2 diabetes
insulin secretion [GO:0030073]
"Highly specific to pancreatic beta cells within the islets of Langerhans."
Pancreatic beta cells
B
Pathway cell contexts by disease
Progressive supranuclear palsy
Major depressive disorder
Alzheimer's disease
Parkinson's disease
Amyotrophic lateral sclerosis
Multiple sclerosis
Systemic lupus erythematosus
Chronic kidney disease
Colorectal cancer
Chronic obstructive pulmonary disease
Obstructive sleep apnea
Crohn's disease
Asthma
Autoimmune thyroiditis
Tuberculosis
Rheumatoid arthritis
Psoriasis
Atrial fibrillation
Heart failure
Hepatocellular carcinoma
Type 2 diabetes
Hypertension
Coronary artery disease
Inflammatory bowel disease
Neurons
Immune cells
Epithelial cells
Fibroblasts
Glial cells
Cardiomyocytes
Hepatocytes
Smooth muscle cells
Intestinal epithelial cells
Pancreatic beta cells
Endothelial cells
Cell-type context
Disease class
Central nervous system / neuropsychiatric
Immune / inflammatory
Cardiometabolic
Respiratory / renal
Epithelial / cancer
Pathway proportion
0
0.5
1.0
C
Disease-level pathway cell-type contexts
Alzheimer's disease
Neurons 280 (90%)
Glial cells 133 (43%)
Hippocampal neurons 111 (36%)
Cortical neurons 109 (35%)
Microglia 73 (24%)
Astrocytes 70 (23%)
Pyramidal neurons 61 (20%)
Excitatory neurons 27 (9%)
Glutamatergic neurons 10 (3%)
Endothelial cells 8 (3%)
Parkinson's disease
Neurons 364 (88%)
Dopaminergic neurons 278 (67%)
Glial cells 123 (30%)
Microglia 51 (12%)
Astrocytes 44 (11%)
Type 2 diabetes
Pancreatic beta cells 361 (47%)
Hepatocytes 258 (34%)
Immune cells 85 (11%)
Endothelial cells 61 (8%)
Macrophages 38 (5%)
Cell-type context group
Neural / glial
Immune
Epithelial
Stromal / vascular
Metabolic
D
Alzheimer's disease: pathway-level cell-type context
endocytosis [GO:0006897] 6 4 4 4 3 3 1 1
plasma membrane [GO:0005886] 7 4 4 2 2 2 1
synapse [GO:0045202] 2 2 2 2
vesicle-mediated transport [GO:0016192] 6 1 1 2 1 1 1
Autophagy - animal [KEGG:04140] 4 2 2 2 3 2
Golgi apparatus [GO:0005794] 4 3 2
Protein processing in ER [KEGG:04141] 4 2 2 4 2 3 4 1
cell adhesion molecule binding [GO:0050839] 4 3 3 2 1 1 3 1
glutamatergic synapse [GO:0098978] 2 2 2 2
lysosome [GO:0005764] 4 1 1 3 3 1
postsynaptic density [GO:0014069] 4 2 2 2 1
Neurons
Cortical neurons
Hippocampal neurons
Glial cells
Pyramidal neurons
Microglia
Astrocytes
Excitatory neurons
Glutamatergic neurons
Endothelial cells
Cell-type context
Tag frequency
1
7

Figure 4